



EGR 489WI SENIOR DESIGN PROJECT

A PROJECT REPORT
PREPARED IN PARTIAL FULFILLMENT OF THE
REQUIREMENT FOR THE DEGREE OF
BACHELOR OF SCIENCE
IN
ENGINEERING TECHNOLOGY

HIGHLAND PLASTICS: THERMAL PERFORMANCE EVALUATION REPORT

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ETHICS STATEMENT AND SIGNATURES

A team consisting of the individuals listed below solely prepared the work submitted in this report and it is original. Excerpts from others' work have been clearly identified, their work acknowledged within the text and listed in the list of references. All of the engineering drawings, computer programs, formulations, design work, prototype development and testing reported in this document are also original and prepared by the same team of students.

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1. INTRODUCTION

1.1. PROJECT OVERVIEW

Highland Plastics is a plastic extrusion and compounding company based in Shepherd, Michigan. They research and produce custom polymers and composites for their clients. The company is currently developing a flame-retardant composite called Kelvinite that will be able to safely contain electric vehicle (EV) battery fires. Development of this material entails significant testing to determine if safety standards are met. Specifically, Kelvinite must resist deteriorating at a series of high temperatures for set periods of time. The current industry standard for efficiently testing these materials is a torch and grit (TaG) test. Purchasing the premade equipment to conduct this test is very expensive, so Highland Plastics has fabricated their own testing equipment. Currently, the test is completely manual and lacks features that would allow for automatic data collection and analysis, better reproducibility, and increased safety. To solve this problem, the team will replace the current testbench with one that is fully automated. In addition to this, the new testbench will more closely follow the UL 2596 standard for the Torch and Grit test and will be modular enough to allow for future upgrades to be installed easily.

The scope of this project encompasses everything required to fully automate Highland Plastics' torch and grit test and add any required safety features. The project must also comply with UL 2596 standards for TaG testing, the only exception being that Highland Plastics has chosen not to use grit to improve test reproducibility. Throughout the course of this project, a PID-controlled air and gas system will be designed and fabricated to control the torch flame, and computer-based software will be developed to control testing parameters, collect relevant data, and display or export data trends. A microcontroller will also be used to control all actuators and compile raw data inputs.

1.2. CUSTOMERS

The primary customer for this project is Highland Plastics, as their specifications and needs drive the design of the project. After the completion of this project, Highland Plastics will take sole ownership of the final product.

The secondary customers involved include Highland Plastic's clients, who may utilize the testing data. Central Michigan University's role as another secondary customer involves supporting the project with resources, mentorship, and academic insight. The student team are additional secondary customers as well, as the group will benefit from the hands-on experience and professional development gained from the project.

1.3. CONSTRAINTS

Highland Plastics is a plastic extrusion and compounding company based in Shepherd, Michigan. They research and produce custom polymers and composites for their clients. The company is currently developing a flame-retardant composite called Kelvinite that will be able to safely contain electric vehicle (EV) battery fires. Development of this material entails significant testing to determine if safety standards are met. Specifically, Kelvinite must resist deteriorating at a series of high temperatures for set periods of time. The current industry standard for efficiently testing these materials is a torch and grit (TaG) test. Purchasing the premade equipment to conduct this test is very expensive, so Highland Plastics has fabricated their own testing equipment. Currently, the test is completely manual and lacks features that would allow for automatic data collection and analysis, better reproducibility, and increased safety. To solve this problem, the team will replace the current testbench with one that is fully automated. In addition to this, the new testbench will

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The project requires a microcontroller that's powerful enough to handle real-time data processing, has many I/O options, and has plenty of documentation. Additionally, any code written for the project must be well documented and easy to debug. The test apparatus will be maintained solely by the current staff, many of whom have no experience with microcontrollers or writing code, so creating code that will be easy to debug will be one of the biggest constraints the student team will have to overcome.

1.4. STAKEHOLDERS

Stakeholders for the project include Highland Plastics, who are the end users of the testing equipment. Central Michigan University is another stakeholder who provides guidance and resources for the project. Another stakeholder is faculty advisor Dr. Mohammed Awad, who supports the student team. The student team is also a stakeholder for the project, as the team directly influenced the design and implementation of resources into the project.

1.5. PROJECT GOALS

This project involves a multitude of goals that are essential to the success of the initiative. The overall goal of this project is to develop an automated test bench that accelerates product development and testing for Highland Plastics by optimizing a flame testing process with a reproducible torch flame. The table must be robust, flame resistant, and include safety features to eliminate any chances of harm. Obtaining accuracy among data collection for each test is another crucial goal for the success of the project. This will be achievable by utilizing a microcontroller and a PID loop to collect and analyze data. By achieving accurate and precise data, the table will become a vital tool for current and future developments within Highland Plastics. Adhering to all design specifications provided by Highland Plastics as well as ensuring that the final product is visually appealing for marketing purposes is also adamant.

1.6. CUSTOMER NEEDS

For this project to be a success, Highland Plastics needs a few key requirements to be met. While some of these needs are small in scope and don't have any impact on the actual testing data, they are still important because of the effect they would have on public perception of the company. For instance, to appear more professional, Highland Plastics wants a new setup to replace their current one that looks professional, has a stopwatch that displays the test time, and will look visually appealing for tests to be filmed and posted online. Additionally, Highland Plastics requires the new setup to give a positive impression of their work area to attract new customers and garner better public opinion. As for the technical aspects of Highland Plastics' needs, the setup must have cold and hot side temperatures prominently displayed, must transmit test data to a computer for analysis, and be easy to fix if any issues were to arise in the future.

1.7. DESIGN TARGETS

Outside of Highland Plastics' needs, there are a few target specifications that must be met to comply with UL-2596 and meet the needs of Highland Plastics. Arguably, the most important key specification is that the burner must reliably achieve and hold a 1200 °C flame (± 50 °C). Since the tests Highland Plastics wants to run completely revolve around the temperature being applied to the composite they are testing, having a flame that can produce a consistent temperature is crucial. Besides having a reliable flame temperature, the setup also needs to have a sample holder that can fit a 200x200 mm piece of composite; this will give Highland Plastics the freedom to experiment with different composites while not compromising on the test quality. The last key specification is that the software the team will write must receive data from a microcontroller and output the test time and hot and cold side temperatures. Moreover, this data must then be sent to a computer for analysis to be done in the future.

2. LITERATURE REVIEW

2.1. UL STANDARDS

The main standard used in the UL Solutions reports is UL 2596, Test Method for Thermal and Mechanical Performance of Battery Enclosure Materials. This standard includes two main test methods, the Torch and Grit Test and the Battery Enclosure Thermal Runaway Test. While these two test methods are similar, the Battery Enclosure Thermal Runaway Test is more realistic since it uses actual lithium-ion cells that are forced into thermal runaway. Outside of these two tests included in the UL 2596 standard, the UL Solutions documents also refer to a few international standards: IEC 60695-4, fire hazard testing, and ISO 13943, fire safety vocabulary. All these standards and codes, while they cover different aspects of testing, combine to provide a repeatable, safe, and effective method for testing the performance of battery enclosure materials.

2.2. FUEL AND AIR STANDARDS

The main standard that relates to the torch system is the National Fire Protection Association (NFPA) 54. NFPA 54 offers criteria for installing and operating gas piping and equipment. Since our system will be utilizing methane gas and oxygen flowing through piping to the torch to perform the tests included in UL 2596, following the guidelines of this standard will allow safe installation and operation of the torch.

2.3. SAFETY CRITICAL PROGRAMMING STANDARDS

IEC 61508 outlines a methodology for assessing the amount of safety required for a project and then considerations depending on that safety level. Our project falls under the category of general production software, which is a level 3 SIL. This level advises using a structured control flow, limiting pointers, recursion, and interrupts, and avoiding implicit type casting.

2.4. ANALYSIS

The standards outlined above serve as a basis for the design of this project. UL 2596 requires that specific parts and materials are used, such as the type of burner and gas. This standard also requires the use of features like an actuating preheat barrier. NFPA 54 restricts the methods and materials used during the installation of our gas system. As a result, our final gas system design must be based on these criteria. Finally, IEC 61508 provides a general guideline on how to structure the code for our data acquisition software.

3. METHODOLOGY

3.1. DESIGN CONCEPTS

3.1.1. BASIC DESIGN: “MODULAR TABLE AND DATA COLLECTION”

This design concept involves the use of a modular table design that utilizes holes for mounting testing equipment. This is done to future-proof the testing table for evolving testing standards. As testing standards may change regarding physical requirements, the table can be adjusted to fulfill these needs. This testing table also has two mounts for air and methane fuel gas canisters. This is significant because mounting the tanks to the table allows for easy maneuverability of the table. The gas tanks are attached to a regulator, control valve, and a torch specified by UL2596. These components will all be mounted to the table as well. The fixture holding the testing material will sit attached to a sliding rail system for maneuverability across from the torch. To allow the torch flame to fully heat before burning the test piece, an electronically enabled stopper will sit between the test piece and the torch, automatically moving out of the way as the torch flame reaches its desired temperature. Thermocouples are attached to a computer system, stopper, and the testing piece will allow for the stopper to move, and data from the test to be automatically recorded to a data logging system on the computer.

3.1.2. DESIGN 2: “BASIC + PID LOOP AND PREDICTIVE OUTPUTS”

The traditional method of reaching a target temperature with a torch involves monitoring the flame temperature with a thermocouple and adjusting the heat input manually or through a PID loop. This approach requires a premium thermocouple rated above 1200 °C and a mechanized system to move components, both of which add cost and complexity. These pain points can be alleviated by using predictive control in the PID loop based on gas flow. The amounts of oxygen and methane entering the torch can be used to estimate the heat output of the reaction, providing a reliable temperature estimate. With this method, no mechanical movement is necessary, since the system releases only the amount of gas required to reach the target temperature quickly, imparting negligible heat rise to the test piece.

3.1.3. DESIGN 3: “BASIC + PID LOOP, PREHEAT BARRIER, AND THERMOCOUPLES”

This design concept uses every aspect of the UL 2596 TaG test apart from the grit system. These aspects include a barrier between the torch and sample that would be automatically removed when the torch has reached a desired temperature, allowing for nearly instantaneous contact of the sample with a fully heated flame. This would allow the test to be more accurate to an actual thermal runaway event, where high temperatures could build very quickly rather than over several minutes. An automated system involving some type of linear actuation will be required to incorporate this concept into the final build. Alternatively, a manual version of this concept could be used for cost savings but would need to keep the user safe from the burner flame.

2-3 type K thermocouples rated to 1300°C would be used in this design as well. This would allow the system to detect when the burner has reached the desired temperature and automatically move the preheat barrier out of the way. Additional thermocouples would be placed on the hot and cold sides of the sample for data logging and to feed data into the PID control loop to maintain a steady flame temperature. This concept also utilizes adjustable modules that would allow for greater test adaptability and an automated data collection system.

3.2. CONCEPT SELECTION

Based on the given target specifications, which call for a burner to reliably achieve and hold a 1200°C flame, have a sample holder that can fit a 200x200 mm piece of composite, and software

that can receive data in real-time from a microcontroller, the concept that best fits the target specifications is Design Concept 3. By using a PID loop, preheat barrier, and thermocouples, the target specifications can reliably be achieved while also providing incredibly accurate data due to the use of thermocouples. Moreover, by utilizing a preheat barrier, it can be assured that the torch is only being used on the test composite when a target temperature has been reached. The instant exposure to a high temperature instead of a slowly rising temperature will also more closely resemble the effects of thermal runaway. Even though this design is more expensive than the other design concepts, the additional cost will result in quality improvements to the data collected.

3.3. PROPOSED PROTOTYPE DESIGN

The proposed prototype selected for development is Concept 3, which incorporates a modular table, data collection system, PID control loop, preheat barrier, and thermocouples. The physical design can be seen in Figure 1, and uses accurate measurements as well as available CAD data. The wiring diagram shown in Figure 2 provides a rough overview of how each component is being powered, and which components require control signals. Figure 3 is a data processing diagram, showing how the captured data is parsed and collated into usable forms.

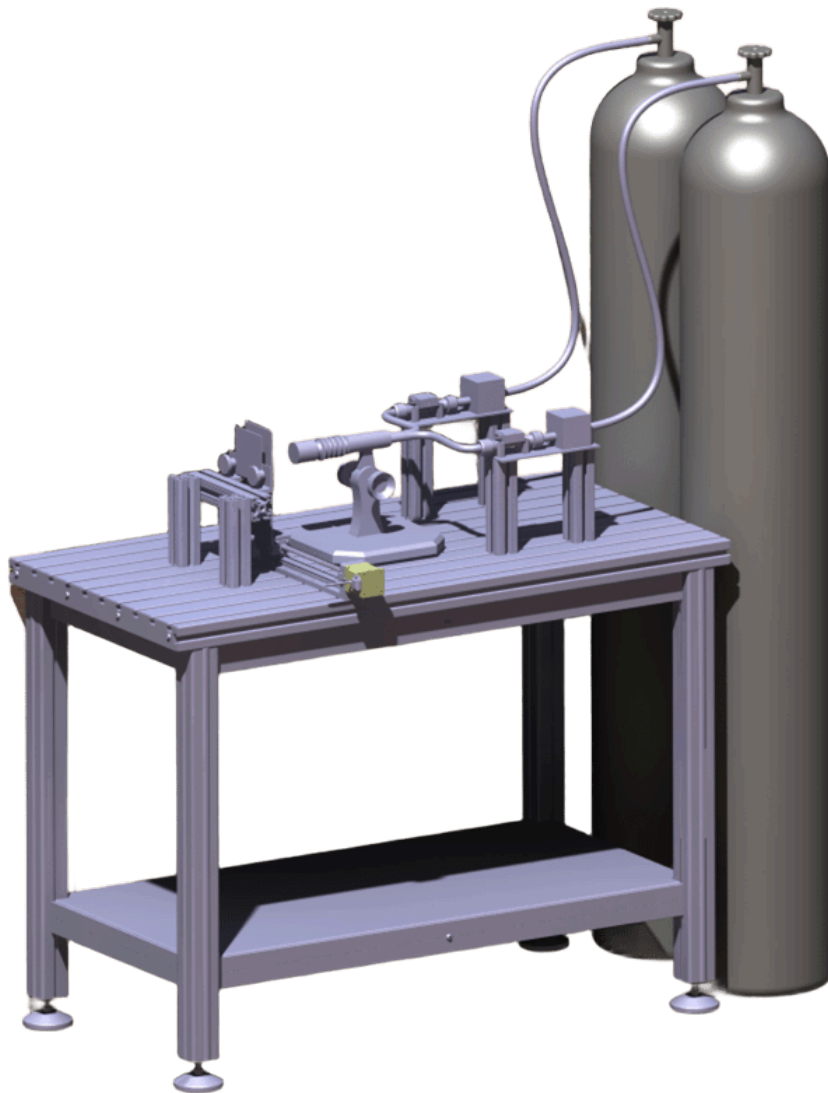


Figure 1: Prototype CAD Model.

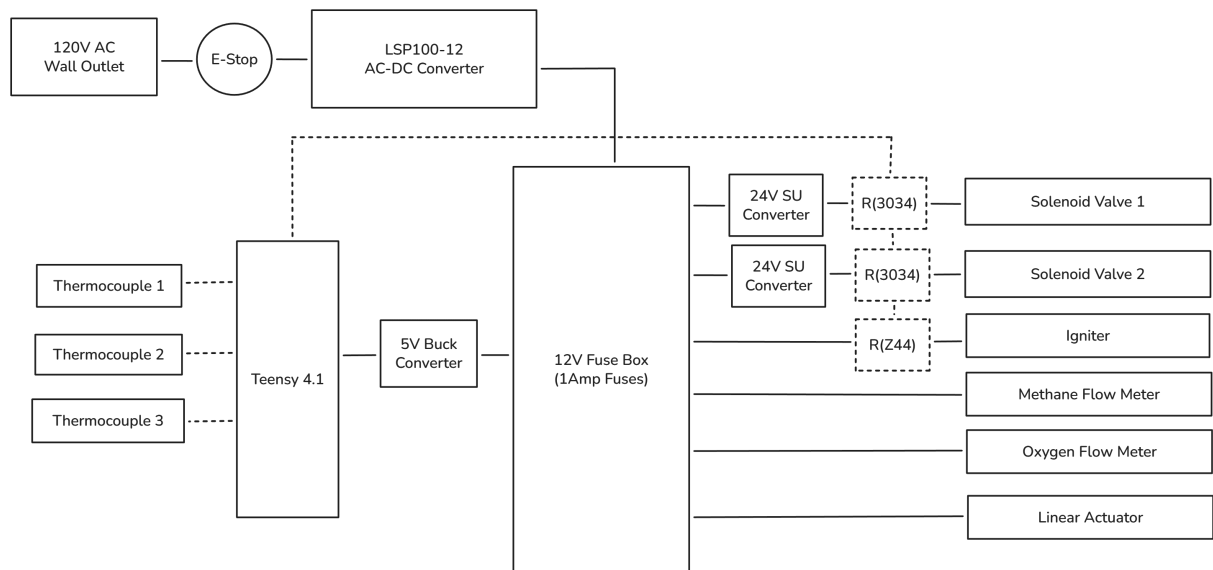


Figure 2: Electrical Wiring Diagram.

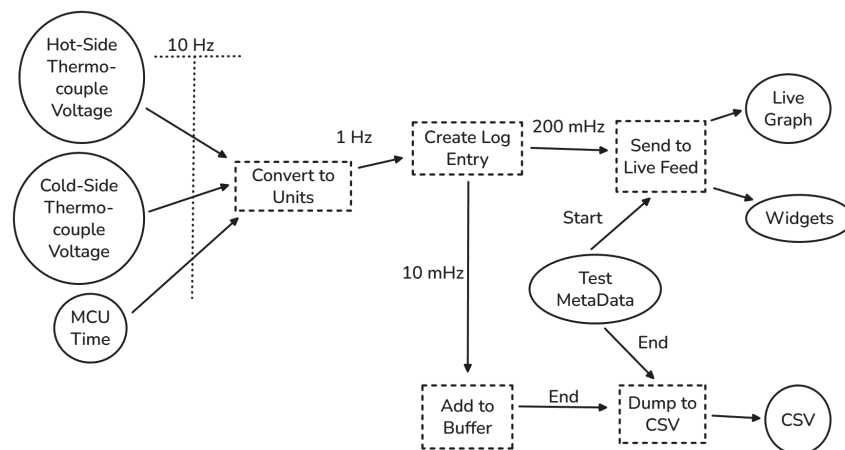


Figure 3: Data Processing Diagram.

3.4. TIMELINE

- October 8-14: Complete first draft of chapters 1-3
- October 14-21: Finish draft for the budget and bill of materials. Additionally, establish space requirements and a project management schedule.
- October 22-28: Finalize supplier list, bill of materials, and management timeline.
- October 29-November 4: Review all work and make any changes needed. Contact Ben and ensure that he's on the same page as the team with the project plans for the second semester.
- November 5-11: Create presentation slides and prepare for presentation.
- November 12-18: Presentations and presentation of peer review.
- November 19-25: Review suggested changes, update project proposal, submit materials order with approval from Dr. Awad.
- November 26-December 10: Finalize all work and submit the final version of the completed proposal. If all work is completed before the deadline, begin to create a general plan for the next semester/building phase.

3.5. BREAKDOWN AND DISTRIBUTION OF WORK

The project is segmented into technical tasks with respective duties for each team member in preparation for development. Matthew Coeling, Ethan Grentz, and Robert Alm will be responsible for the PID Loop and System Wiring, collaboratively designing, integrating, and constructing the control system with emphasis on functionality and wire organization. Connor Bendele will oversee microcontroller setup and hardware integration, including programming, hardware compatibility verification, and coordinating with Robert to enable software integration. Robert Alm will also handle software development, tasked with creating the desktop application, version control, and system testing and validation guidance. Sam Hass and Elijah Dawson will be the project leads for Fuel System and Fixture Design. Sam will manage CAD modeling, physical assembly, and coordination with Highland Plastics for the team. Elijah will be involved in design and assembly tasks and perform flow rate calculations.

3.6. BILL OF MATERIALS

Description	Quantity	Price	Item Number	Source
T-Slot Fixture Table	1	\$1,786.15	7710N14	McMaster-Carr
Cylinder Rack	1	\$160.84	22832T64	McMaster-Carr
Gas Regulator	2	NA	NA	Highland Plastics
Teensy 4.1 MCU	1	\$31.50	DEV-16771	Sparkfun Electronics
Solenoid Flow-Adjustment Valve	2	\$87.92	2555N12	McMaster-Carr
Safety Stop	1	\$40.00	BC66ESB/P	ValveShop
Type K Thermocouple	3	\$43.68	5FYH1	Grainger
Linear Actuator	1	\$172.24	LMD17S19WF25-200	SterppperOnline
Spring Clamp Work Holder	1	\$100.15	5007A18	McMaster-Carr

Hex Head Screws 50 Pack	1	\$8.57	92979A122	McMaster-Carr
Socket Head Screws 100 pack	1	\$13.23	91290A115	McMaster-Carr
90deg 6061 Aluminum	1	\$23.53	982K135	McMaster-Carr
304 Stainless Steel Sheet	1	\$28.71	6620K181	McMaster-Carr
Steel Flange Nuts 100 pack	1	\$10.31	94842A101	McMaster-Carr
1" Threaded Rod	2	\$1.51	8758A305	McMaster-Carr
1.5" Threaded Rod	4	\$1.90	98758A308	McMaster-Carr
Setup Clamp	6	\$14.50	84945A47	McMaster-Carr
Threaded T-Slot Nut	6	\$3.83	94750A588	McMaster-Carr
12-24V DC Step-Up Converter	2	\$18.49	EKYLIN-DC12-24V-5A	Amazon
IRLB3034 Transformer	1	\$17.99	IRLB3034PBF	Amazon
Motor Driver	1	\$9.99	B0C5JCF5RS	Amazon
Spark Igniter	1	\$27.03	PRO622414105	
Brass Manual Valve	2	\$14.55	47865K23	McMaster-Carr
3ft Brass Pipe	2	\$35.80	4501K52	McMaster-Carr
12V Fuse Box Blade	1	\$16.99	B0B9BVCK7M	Amazon
120AC - 12VDC Converter	1	\$27.50	LSP-100-12	Mouser Electronics
1A Fuses 25 pack	1	\$5.99	B01H9XALY0	Amazon
24 AWG Shielded Wire	1	\$12.00	B0CJTWLQV4	Amazon
20 AWG Shielded Wire	1	\$8.99	B0FH4ZWCJ2	Amazon
16 AWG Shielded Wire	1	\$28.98	B0BPQCTRTX	Amazon
IRLZ44N MOSFET Transistors 10 pack	1	\$9.99	B0CBKH4XGL	Amazon
12-5V Buck Converter 2 pack	1	\$14.99	B0DQKFZ2BF	Amazon
Total Cost: \$3120.87				